

Rapid Exhumation of High-Pressure Metamorphic Rocks in Kythera-Peloponnese (Greece) Revealed by Apatite (U-Th)/He Thermochronology

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The Journal of Geology (2014)

DOI: <https://doi.org/10.1086/675908>

Abstract. Apatite (U-Th)/He and apatite fission-track cooling ages from high-pressure (HP) metamorphic rocks in the South Aegean forearc along the Kythera Strait (Kythera and southeastern Peloponnese) suggest rapid exhumation during Middle to Late Miocene times. HP rocks from the Kythera Strait exhibit two perpendicular extensional structures. Samples with arc-normal or arc-parallel extensional fabrics yielded apatite (U-Th)/He cooling ages of ~8–10 Ma. The more precise (U-Th)/He ages are consistent with previously reported apatite and zircon fission-track ages obtained from the same rock units. The less precise, but still accurate, fission-track data provided a reference for (U-Th)/He dating on relatively young uranium-poor apatites. The combined data, therefore, precisely and accurately constrain the exhumation history of HP rocks and indicate that localized arc-parallel extension in the Kythera Strait was active for an ~5 m.yr. interval during the regional arc expansion.

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